PROBLEM FRAMING BRIEF

NeuroTechh

Appointment Coordination Platform for patients and clinics

1 November 2026
MAL Hackathon 4
HealthTech

The problem. Patients cannot see real availability, and clinics cannot control it reliably: slots are double-promised, cancelled slots go unused, and requests are chased by phone. NeuroTechh is a multi-tenant mobile platform where patients request appointments and clinics confirm them. **Its one non-negotiable promise: a confirmed slot belongs to exactly one patient, and a cancellation returns the slot to everyone, consistently.**

USERS

Patient
MOBILE APP
Finds a doctor by locality, specialisation and filters; requests, tracks, reschedules or cancels; holds a QR pass. Can book for dependents.

Hospital admin
WEB + MOBILE
Sets booking windows and policies; confirms or declines requests; manages doctors and blocked dates.

Doctor
WEB + MOBILE
Controls own availability for new and follow-up visits; posts notes and prescriptions; reviews history the patient has shared.

Front-desk staff
SCANNER APP
Scans the patient's QR pass to verify the booking and check them in.

CORE JOURNEY

1 Browse	2 Request	3 Decide	4 Pass + alerts	5 Visit	6 Follow up
Search by locality, specialty or doctor and see live slots.	Patient requests a slot. Pending requests do not block others.	Clinic confirms one request and declines the rest, or auto-confirms by policy.	SMS, email and push. QR pass is saved offline and emailed.	Staff scan the QR to check in. Doctor posts notes and prescription.	Notes inform the next booking, with rules set by the doctor.

At any point before the visit the patient or clinic can reschedule or cancel. The old QR pass is revoked, the patient and doctor are notified, and the slot reopens to the public.

SCOPE

In scope

- **Phase 1, core loop:** browse, request, confirm or decline, status, cancel, reschedule, QR pass, notifications, admin and doctor tools, payments.
- **Phase 2, records:** patient uploads, prescriptions and notes, consent-based sharing, follow-ups.
- **Phase 3, extras:** nearby chemists and labs, analytics.

Out of scope

- Clinical advice and diagnosis; video consultations; replacing a hospital's own EMR or HIS (integration is a later option).

ASSUMPTIONS

- **Clinics always confirm.** No instant booking; auto-confirm is an optional per-clinic setting.
- **Pending requests hold nothing.** Approving one declines the rest.
- **Hospitals set the rules:** booking window, lead time and cancellation cutoff.
- **Time:** stored in UTC, shown in the clinic's local time.
- **Records are in scope temporarily,** pending compliance review.
- **Money:** patients pay per appointment, hospitals subscribe; payment timing to be confirmed.
- **Sharing across hospitals** happens only with explicit patient consent.

CONSTRAINTS

- **Integrity:** the database is the final authority on slot ownership; Redis locks only reduce contention.
- **Compliance:** global. Designed to DPDP (India), HIPAA and GDPR; core DPDP duties apply from May 2027.
- **Residency:** data is sharded by region and stays in the patient's home region.
- **Privacy:** QR codes, emails and SMS carry no health data; every record access is audited.
- **Platform:** React Native, microservices on Kubernetes, PostgreSQL, Redis, ClickHouse for audit.
- **Scale:** built for millions of patients; the patient's QR pass must work offline.

SUCCESS CONDITIONS

- **Zero double allocations:** no confirmed slot ever has two patients.
- **Consistent views:** a cancelled slot reopens and patient and clinic screens agree within seconds.
- **No orphaned requests:** every request is answered or expires within the clinic's window (default 24 hours), with admin reminders.
- **QR works at the door:** verifies even when the patient is offline, and stops working once cancelled.
- **Auditable:** all record access is logged and shared only with consent.
- **Proven in a pilot:** design-partner hospitals run the full loop before wider rollout.